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FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

**DESIGN AND IMPLEMENTATION OF AN AI-ENABLED
INTEGRATED MULTI-APPLICATION FOR BUSINESS
MANAGEMENT AND QR CODE DISCOVERY: iHakken**

*Internship Report Submitted to the Department of Computer
Engineering, Faculty of Engineering and Technology, University of
Buea, in Partial Fulfilment of the Requirements for the Award of the
Bachelor of Engineering (B.Eng.) Degree in Computer Engineering*

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2025/2026 Academic Year

CERTIFICATION

This report hereby confirms that ABANDA SERGIO ABANDA, with matriculation number FE22A131, a student of the University of Buea, Faculty of Engineering and Technology, Department of Computer Engineering, majoring in Software Engineering, has completed his internship report on the project “**DESIGN AND IMPLEMENTATION OF AN AI-ENABLED INTEGRATED MULTI-APPLICATION FOR BUSINESS MANAGEMENT AND QR CODE DISCOVERY**”, submitted in partial fulfillment of the requirements for the award of the Bachelor of Engineering Degree in Computer Engineering, under the academic supervision of Prof. Elie Fute Tagne and the professional supervision of Mr. Sone Gillis

Professional Supervisor:

Date:

Signature:

Academic Supervisor:

Date:

Signature:

DEDICATION

This work is dedicated to my beloved mother, **Mrs. Abanda Quinta Adong**, whose unwavering love, sacrifices, prayers, and encouragement have been the foundation of my academic and personal journey. Her strength and support continue to inspire me to strive for excellence.

ACKNOWLEDGEMENTS

First and foremost, I give all glory and honor to God Almighty for granting me life, strength, wisdom, and guidance throughout my internship period and for the successful completion of this report.

I am sincerely grateful to the Head of the Department of Computer Engineering and my Academic Supervisor, Prof. Elie Fute T, for his academic guidance, constructive feedback, and encouragement during the preparation of this report. I also extend my heartfelt appreciation to the Faculty of Engineering and Technology, University of Buea, particularly the Dean, Prof. Agbor Dieudonne Agbor, and all my lecturers in the Department of Computer Engineering, for equipping me with the solid theoretical and practical foundation that enabled me to perform effectively during my internship, with special thanks to the lecturers of the Department of Computer Engineering for their guidance and support, which gave me a strong foundation to perform effectively in my professional experience.

I express my profound gratitude to my Professional Supervisor and Founder of Kunshort Company Ltd, Mr. Sone Gillis, for his mentorship, technical guidance, leadership, and continuous support throughout the internship. I am equally thankful to Mr. Mbassi Darlington Mbassi, Operations and Resource Manager, and Akum A. Challand Binda, Financial and Marketing Director of Kunshort Company Ltd, for providing me the opportunity to carry out this professional experience. I sincerely appreciate my team members at Kunshort for their collaboration, teamwork, and knowledge sharing during the development of the iHakken platform, which greatly contributed to my professional growth.

Finally, I wish to express my heartfelt gratitude to my beloved mother, Mrs. Abanda Quinta Adong, and my sister, Ahishia Kabiru Lami, for their unconditional love, sacrifices, prayers, and unwavering moral support throughout my academic journey. I also thank my classmates and friends for their encouragement, cooperation, and motivation, which inspired me to complete this internship and report successfully.

ABSTRACT

This internship report presents the design and implementation of an integrated multi-application platform for business management and QR code discovery known as iHakken. The internship was carried out at Kunshort Company Ltd, a technology solutions and startup-building company, as part of the requirements for the award of the Bachelor of Engineering (B.Eng.) degree in Computer Engineering at the University of Buea. The primary objective of the internship was to apply theoretical knowledge of software engineering to real-world application development while gaining hands-on experience in modern web technologies and professional software development practices. During the internship, I actively contributed to the development of the iHakken platform, which is designed to support hospitality businesses such as restaurants and lodging establishments. The platform consists of an administrative application for business owners and a user-facing application accessed through QR codes by customers.

Key responsibilities included the development of frontend components using React, Next.js, and TypeScript, implementation of assignment management systems, analytics dashboards, QR code generation modules, staff and table management systems, and the integration of AI-powered chat assistants. Additionally, exposure to DevOps practices such as Continuous Integration and Continuous Deployment (CI/CD) using GitHub Actions enhanced my understanding of automated deployment workflows. Despite challenges such as learning new frameworks, navigating a large codebase, power supply issues, and security-related disruptions, effective problem-solving strategies were adopted to ensure successful task completion. Overall, the internship provided valuable technical skills, professional experience, and insight into real-world software engineering, thereby bridging the gap between academic learning and industry practice.

Some Key Words and Their Meanings

- **iHakken:** It is an integrated digital platform developed to help hospitality businesses like restaurants and hotels manage their services efficiently. It includes an admin application for business owners and a user application that customers access through QR codes to browse menus, request services, and interact with AI assistants.
- **Contabo Virtual Private Server (VPS):** A VPS is a remote server that hosts applications on the internet. In this project, the Contabo VPS was used to deploy and run the iHakken applications securely so that users can access them online at any time.
- **React and Next.js:** React is a JavaScript library used to build interactive user interfaces, while Next.js is a framework built on top of React that improves performance through features like server-side rendering. Together, they were used to develop the frontend of the iHakken platform.
- **QR Code Generation:** This refers to the process of creating scannable QR codes that link customers directly to specific services such as restaurant tables, menus, or hotel accommodations. It enables contactless access to services through mobile devices.
- **Continuous Integration and Continuous Deployment (CI/CD):** CI/CD is an automated process that builds, tests, and deploys applications whenever developers update the code. In this project, GitHub Actions was used to automatically deploy updates to the server, reducing manual work and deployment errors.
- **Hospitality Management System:** A digital system designed to manage operations in hospitality businesses such as restaurants and lodging establishments. In this project, iHakken functions as a hospitality management system by integrating service management, analytics, QR access, and AI-powered customer interaction.
- **WebSocket:** WebSocket is a communication technology that keeps a constant connection open between the client and the server. Instead of sending separate requests each time, it allows data to be sent and received instantly in real time. In this project, WebSocket was used to power live chat interactions and real-time service requests within the iHakken user application.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL MEANING
AI	Artificial Intelligence
API	Application Programming Interface
B.Eng.	Bachelor of Engineering
CEO	Chief Executive Officer
CI/CD	Continuous Integration / Continuous Deployment
CV	Curriculum Vitae
DevOps	Development and Operations
DME	DME Systems
ES6	ECMAScript 6
FET	Faculty of Engineering and Technology
GCE	General Certificate of Education
GSM	Global System for Mobile Communications
HTML	HyperText Markup Language
ITU	International Telecommunication Union
RAG	Retrieval-Augmented Generation
RTK Query	Redux Toolkit Query
UI	User Interface
UX	User Experience
VPS	Virtual Private Server
JS	JavaScript
WS	WebSocket
SEO	Search Engine Optimization

CHAPTER 1: GENERAL INTRODUCTION

1.1 Definition of Internship

An internship is a period of work experience offered by an organization for a limited time. While it is a key part of the journey for students, internships are also open to graduates or individuals looking to gain new skills in a different field. It acts as a bridge between classroom learning and the actual work done in the professional world. It facilitates the transition from theoretical understanding to the execution of real tasks in a professional environment. Through an internship, an individual learns how a workplace operates and builds the practical skills needed for a successful career.

1.2 Goals and Objectives of Internship

The main reason for an internship is to give someone the chance to practice what they have learned in a real-life setting. Some of the main goals include:

- **Practical Application:** To use knowledge gained from studies to solve real problems in a work environment.
- **Skill Development:** To learn how to work in a team, communicate professionally with colleagues, and follow workplace discipline.
- **Industry Orientation:** To see how companies in a specific field operate and understand the tools and technologies they use every day.
- **Career Preparation:** To get ready for the job market by learning from experienced professionals and mentors.

1.3 Posting and Application for Internship

The process of getting an internship usually starts with finding an organization that matches one's interests or field of study. Generally, an applicant submits a formal request, often including an introductory letter, curriculum vitae (CV), and other academic documents. The organization then reviews these to see if the applicant is a good fit. This process may involve an interview or a technical test. Once accepted, the person is assigned to a specific department where they work under the guidance of a supervisor for the duration of the program.

The journey to securing my internship was a selective process aimed at finding an environment that offered both technical structure and alignment with my career interests. Before joining Kunshort Company Ltd, I explored opportunities at two other organizations.

My first placement was at DobreTech, located at the University of Buea IT Center. While the environment was professional and well-equipped, the organization did not have a structured

plan specifically tailored for software engineering interns at the time. After staying there for two weeks and realizing I was not gaining the intended practical growth, I decided to seek a more challenging environment.

Next, I applied to the Orange Digital Center at the University of Buea. I secured a spot by sending a formal email to the coordinator. However, I found the program to be overcrowded, which limited personal mentorship. Furthermore, the available path focused heavily on Data Science, which did not align with my passion for Software Engineering and application development.

Despite being accepted at both previous locations, I chose to keep searching for the right fit. I discovered Kunshort Company Ltd through a social media post shared by a friend on WhatsApp, stating that the company was looking for interns. After consulting with another friend who provided more details about the company's Startup Builder model and professional culture, I became highly interested.

I submitted my application via email, including the following documents:

- An internship request letter.
- Photocopies of my University Transcripts.
- GCE Ordinary and Advanced Level Certificates.
- An introductory letter from the Faculty of Engineering and Technology (FET).

I was accepted into the program and officially posted to the Software Development Department. I chose Kunshort because it offered a clear learning path in Fullstack Development, a structured training program, and the opportunity to work on innovative products like iHakken.

1.4 Presentation of the Company: Kunshort Company Ltd.

1.4.1 Name and Location of the Company

Kunshort Inc. is a technology solutions company and startup builder. While it is officially incorporated in Delaware, United States, its main office and operational base are located in Buea, South West Region Cameroon.

1.4.2 About Kunshort Company Ltd.

Kunshort is a budget friendly startup builder and app development company that transforms ideas into scalable digital solutions. The company specializes in developing software applications across all platforms for clients worldwide. Kunshort is led by a team of experienced professionals focused on delivering reliable software's that supports businesses in global markets. The office operates from 8:00 AM to 6:00 PM, and they can be reached at contact@kunshort.com.

1.4.3 Vision, Mission, and Core Values

The mission of Kunshort is to empower entrepreneurs and organizations with digital solutions that are innovative, affordable, and scalable. They aim to turn bold ideas into thriving businesses through technology. The company's work is guided by four core values:

- **Innovation:** Building smart, practical, and impactful solutions.
- **Partnership:** Growing alongside clients rather than just serving them.
- **Integrity:** Maintaining transparency and accountability in all projects.
- **Community:** Nurturing local talent and uplifting the technology ecosystem.

1.4.4 Services Rendered

Kunshort Inc. provides a wide range of IT services to help businesses grow:

- **Web and Mobile Development:** Crafting responsive websites and mobile apps using frameworks like **React Native** and **Next.js**.
- **UI/UX Design:** Creating intuitive and visually stunning designs for a great user experience.
- **Data Analytics and Artificial Intelligence:** Using AI to help companies understand data and predict future trends.
- **Cloud and Cyber security:** Ensuring that digital platforms are secure and reliable.
- **Education and Training:** Providing hands-on skills in coding, web design, and IT management.
- **Freelance Services:** Offering expert technical support for branding and software creation.

1.4.5 Organizational Structure

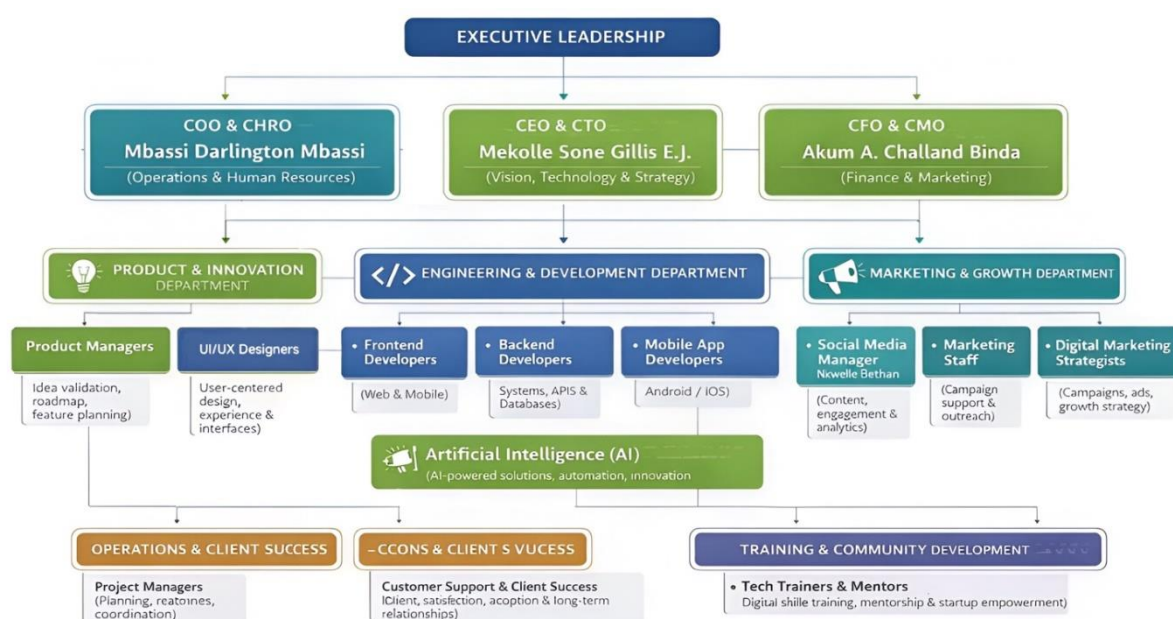


Figure 1: Kunshort organization structure

1.4.6 Technologies Used

Kunshort Company Ltd uses a modern set of tools to ensure their applications are fast, secure, and professional. The following technologies were used during the development of various projects:

- **Next.js:** A web framework used to build fast, user-friendly websites that load quickly and perform well on search engines.
- **React Native:** A tool used to build mobile applications for both Android and iOS using a single codebase.
- **Python:** A versatile programming language used for writing the logic and "brains" of the backend systems.
- **Django:** A powerful Python-based framework that helps developers build secure and scalable backends very quickly.
- **PostgreSQL:** A highly reliable database system used to store and manage complex company and user data securely.
- **Docker:** A technology used to package software into "containers," ensuring it runs perfectly on any computer or server.
- **JavaScript / TypeScript:** The primary languages used for writing the interactive parts of the web and mobile applications.
- **Git & GitHub:** Tools used for version control, allowing the team to save their work and collaborate on code without errors.

CHAPTER 2: INTERNSHIP ACTIVITIES

2.1 First Day of Internship: Induction and Orientation

My internship at Kunshort Company Ltd officially commenced on 15th August 2025, with my arrival at the office before 9:00 AM. The day was structured to provide a comprehensive introduction to the company's vision, leadership, and technical expectations.

2.1.1 Interaction with the CEO and Company Briefing

The induction began with an introductory session led by Mr. Sone Gillis, the CEO and my Professional Supervisor. He opened the floor by introducing himself and invited each intern to do the same, stating our names and respective fields of study. Following the introductions, Mr. Sone provided an in depth briefing on the company's dual business model. He explained that Kunshort focuses on:

1. **Client-Based Projects:** Developing tailor-made software solutions for external clients.
2. **Internal Product Development:** Building and deploying proprietary software that runs directly under the company.

He emphasized the goal of the company as a Startup Builder and introduced us to the technical frameworks that drive their innovation, specifically highlighting the tools we would be mastering during our stay.

2.1.2 Meeting the Leadership Team (The Board)

Later in the morning, the rest of the company's leadership joined the session. In total, we met with four key members, including the CEO. Each representative introduced themselves, explained their specific roles within the organization, and offered words of encouragement. This interaction was highly motivating, as it gave us a clear sense of the professional standards and the supportive culture within the firm.

2.1.3 Project Roadmap and Technical Orientation

To give us a vision of our future contributions, the leadership presented ten (10) potential project topics that the company has in its pipeline. Each project was explained in detail, and we were encouraged to share our thoughts and understanding of these concepts. We were informed that the first two months would be dedicated to an Intensive Training Phase. This period was designed to ensure we mastered the company's frameworks and development tools before being integrated into live project teams.

2.1.4 First Technical Task: Modern JavaScript (ES6+)

Before the day concluded, we were assigned our first official task. This was a Day 1 Refresh on **Modern JavaScript (ES6+)**, which is the foundation of the company's tech stack. The task covered several critical concepts:

- **Variable Scoping:** Understanding let and const.
- **Modern Syntax:** Arrow functions, Template literals, and Destructuring.
- **Data Handling:** Spread and Rest operators.
- **Asynchronous Programming:** Working with Promises and Async/wait.
- **Modularization:** Understanding the Import/Export system.

The practical part of this task required us to convert a traditional "Vanilla JS" application into modern ES6 syntax. We immediately began our studies, marking a productive and inspiring start to the internship.

2.2 Technical Supervisor

During the course of my internship, I was placed under the direct technical supervision of Engr. Sone Gillis, who serves as the CEO and Lead Software Engineer at Kunshort Company Ltd. Having a supervisor with such an extensive and diverse background in the global tech industry was instrumental to my professional growth.

2.2.1 Academic and Professional Background

Engr. Sone Gillis is an Electrical Engineering graduate with a major in Telecommunications from the Faculty of Engineering and Technology(FET) in the University of Buea. His journey in technology began with a strong foundation in hardware (electronic circuits and microcontrollers) before transitioning into high-level software engineering.

His professional career is marked by significant roles in both local and international organizations, including:

- **DME Systems (formerly Softtech International):** Where he rose from a junior to a Mid-Level Software Engineer, working on complex sports betting applications.
- **International Telecommunications Union (ITU):** Where he served as a Software Engineer Consultant, optimizing global conferencing applications.
- **IBL Education:** Where he served as a Software Engineer, building educational platforms for world-renowned clients such as Tesla, NVIDIA, and Western Governors University

2.2.2 Mentorship Style and Influence

As a supervisor, Engr. Sone brought his experience as a Technical Team Lead to our daily interactions. He emphasized not just "writing code," but writing clean, optimized, and efficient code. His background in solving local problems, such as building Hospital Management Systems and GSM networks for remote areas inspired us to look at software as a tool for community impact.

2.3 Department of Posting

During my internship at Kunshort Company Ltd, I was officially posted to the Frontend Development Unit within the Software Engineering Department. This unit is responsible for building the "client-side" of applications, the part that users see and interact with. My role was to ensure that complex backend data was presented through intuitive, responsive, and high-performance interfaces.

2.3.1 The Software Development Department

This department serves as the "engine room" of the company, where digital concepts are transformed into functional products. Being posted here allowed me to work directly on the Frontend architectures of live applications. The department follows an Agile Development methodology, which involves:

- **Collaborative Coding:** Working alongside other developers to build different modules of the same application.
- **Peer Code Reviews:** Sharing our code with senior engineers (like Engr. Sone Gillis) to ensure it meets the company's quality and security standards.
- **Daily Stand-ups:** Brief meetings to discuss the progress of tasks, challenges faced, and the goals for the day.
- **Progress Tracking:** To maintain accountability, I was responsible for submitting formal **daily reports**. These reports detailed the tasks completed during the day and outlined the roadmap for the following day's objectives. This practice helped me develop professional discipline and improved my ability to estimate development timelines accurately.

2.4 Introduction to Ihakken Platform

Ihakken is a comprehensive digital platform designed specifically for businesses operating in the hospitality industry, focusing on restaurants and lodging establishments. The platform consists of two main applications that work together to provide a complete solution: the

Ihakken-admin application (administrative interface) and the Ihakken-user application (customer-facing interface).

2.4.1 The Ihakken-admin application

The Ihakken-admin application serves as the administrative backend that empowers business owners and administrators to manage their entire operation through a unified, user-friendly web interface. The application provides a complete suite of management tools that streamline various aspects of hospitality business operations. It enables businesses to manage their multi-location operations, assign menu items and accommodations to different service locations, configure AI-powered customer service assistants, generate QR codes for contactless service delivery, and analyze business performance through comprehensive analytics.

The platform is built using modern web technologies including React with TypeScript for the frontend, ensuring type safety and maintainability. The application follows a component-based architecture that promotes code reusability and consistency across different modules. State management is handled through Redux Toolkit Query (RTK Query), which provides efficient API communication, automatic caching, and real-time data synchronization.

The Ihakken-admin application supports two primary business types: restaurants and lodging establishments. For restaurants, the platform facilitates menu management, table assignments, and customization options. For lodging businesses, it provides floor management, accommodation assignment, availability tracking, and amenity configuration.

One of the key features of the admin application is its ability to generate customizable QR codes that link customers directly to specific services, tables, or accommodations. These QR codes enable contactless interactions, allowing customers to view menus, request services, and access information about accommodations through their mobile devices. The application also integrates AI-powered Chabot functionality that can be configured for different service locations or staff units. These AI assistants can handle customer inquiries, provide information about services, and assist with various requests, enhancing the overall customer experience while reducing the workload on staff.

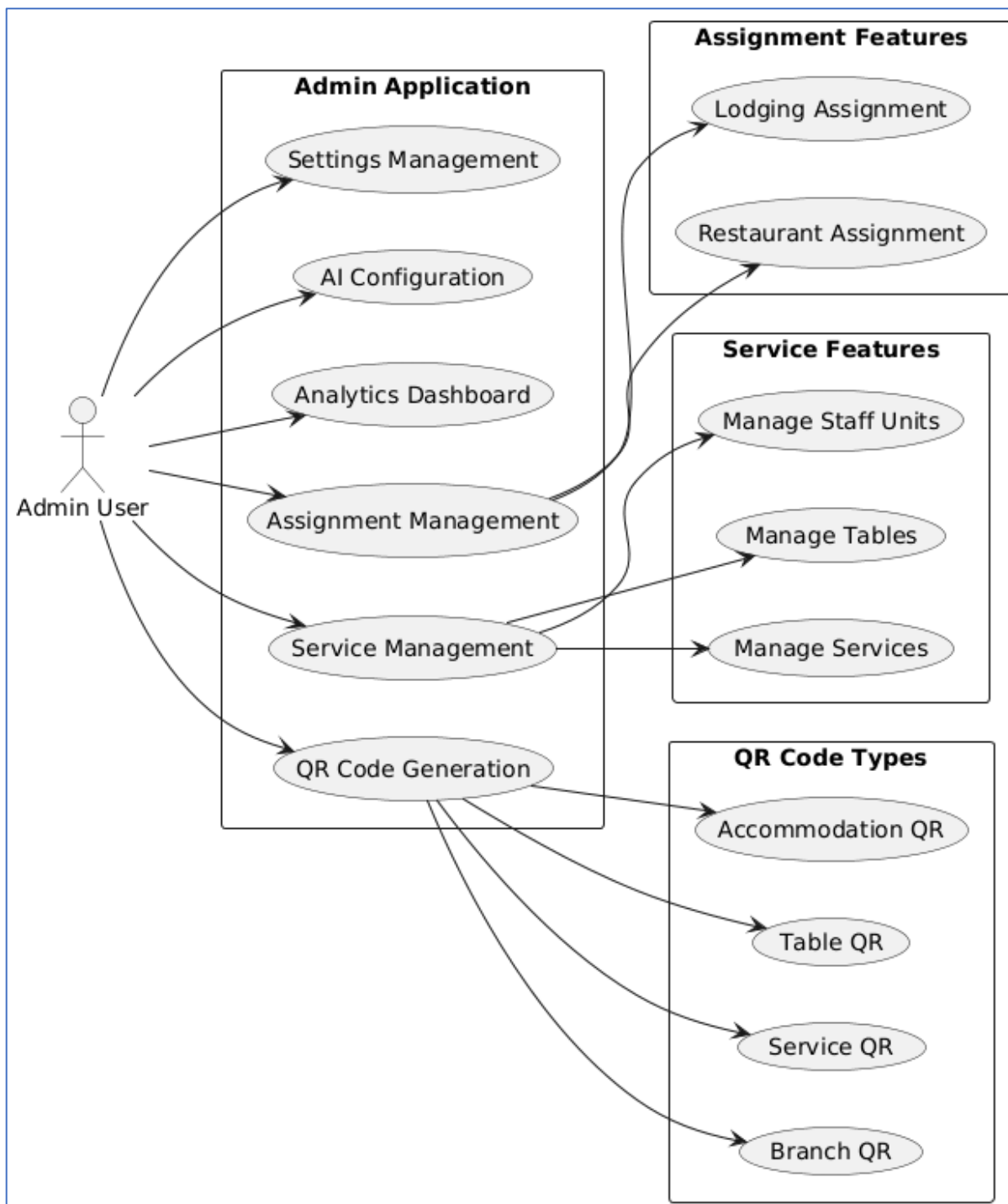


Figure 2: Use Case Diagram Showing Overall Admin-Application

2.4.2 Ihakken-User Application

The Ihakken-user application is the customer-facing web application that provides an intuitive and seamless experience for end-users (customers) interacting with hospitality businesses. This application is accessed primarily through QR codes scanned from various touch points within restaurants or lodging establishments.

2.4.1 Core Functionalities:

QR Code Integration: Customers scan QR codes from branch, tables, accommodation, or service displays, which automatically decode and route them to the appropriate service interface

Restaurant Experience: Customers can browse menus, view menu items by categories, search for specific dishes, view detailed item information, customize orders, and request waiter assistance directly from their tables

Lodging Experience: Guests can explore available accommodations, view room details with images and amenities, check pricing, and request various services through staff units

AI Chat Assistant: Real-time AI-powered chat interface that allows customers to interact with AI assistants, ask questions about menu items, accommodations, or services, and receive instant responses

Service Requests: For restaurants, customers can request for waiters directly from their table. For lodging, guests can initiate service calls to different staff units (front desk, room service, maintenance, etc.)

Real-time Communication: WebSocket-based communication for live chat interactions and service calls

2.4.2 Technical Architecture:

The user application is built with Next.js (React framework) and TypeScript, providing server-side rendering capabilities and optimal performance. The application handles QR code payload decoding, session management through localStorage, and real-time WebSocket connections for chat and service calls. The UI is designed to be mobile-first, responsive, and accessible, ensuring a smooth experience across all devices.

The seamless integration between the admin and user applications ensures that any configuration made by business owners in the admin app is immediately reflected in the customer experience, creating a unified ecosystem for hospitality service management.

Throughout my internship, I was responsible for developing critical components and features in both applications, working collaboratively with a team of developers to build a robust and scalable platform that meets the diverse needs of hospitality businesses and their customers.

2.5 Tasks Assigned to Me

During my internship period, I was assigned several critical development tasks as part of building the Ihakken admin application. The primary focus of my work was developing the

frontend components and user interfaces that would enable business owners to manage their restaurant and lodging services efficiently. The main tasks assigned to me included:

2.5.1 Assignment Management System

I was responsible for developing a comprehensive assignment interface that allows businesses to assign menu items, categories, and accommodations to their services. This system was divided into two main modules:

2.5.1.1 Restaurant Assignment Module

I developed a drag-and-drop interface that enables restaurant owners to assign menu items and categories to specific branch services. The module includes functionality for customizing menu items with ingredients, toppings, complements, and add-ons, allowing each service location to have unique pricing and availability settings. The system supports day of week Availability configuration and custom pricing for menu items assigned to different branches.

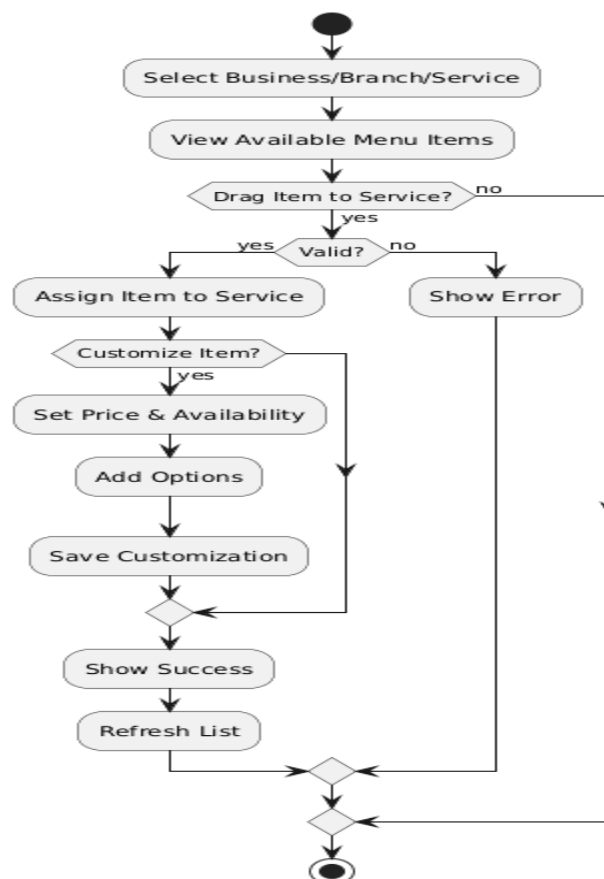


Figure 3: Activity Diagram for Restaurant Menu Item Assignment Process

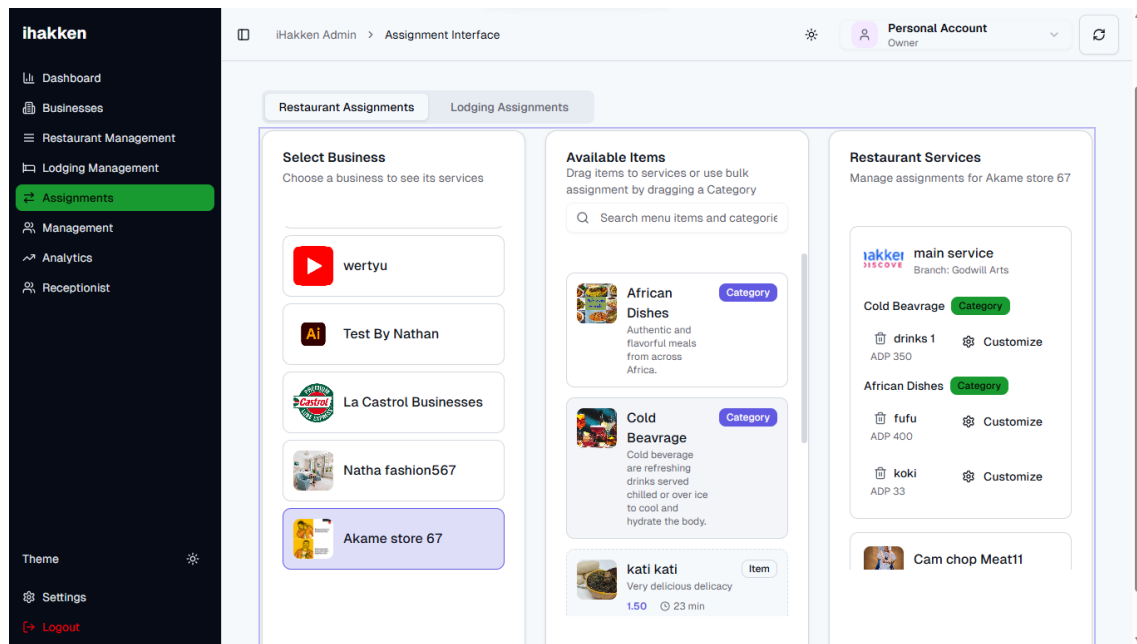


Figure 4: User Interface Screenshot Showing Restaurant Assignment Interface

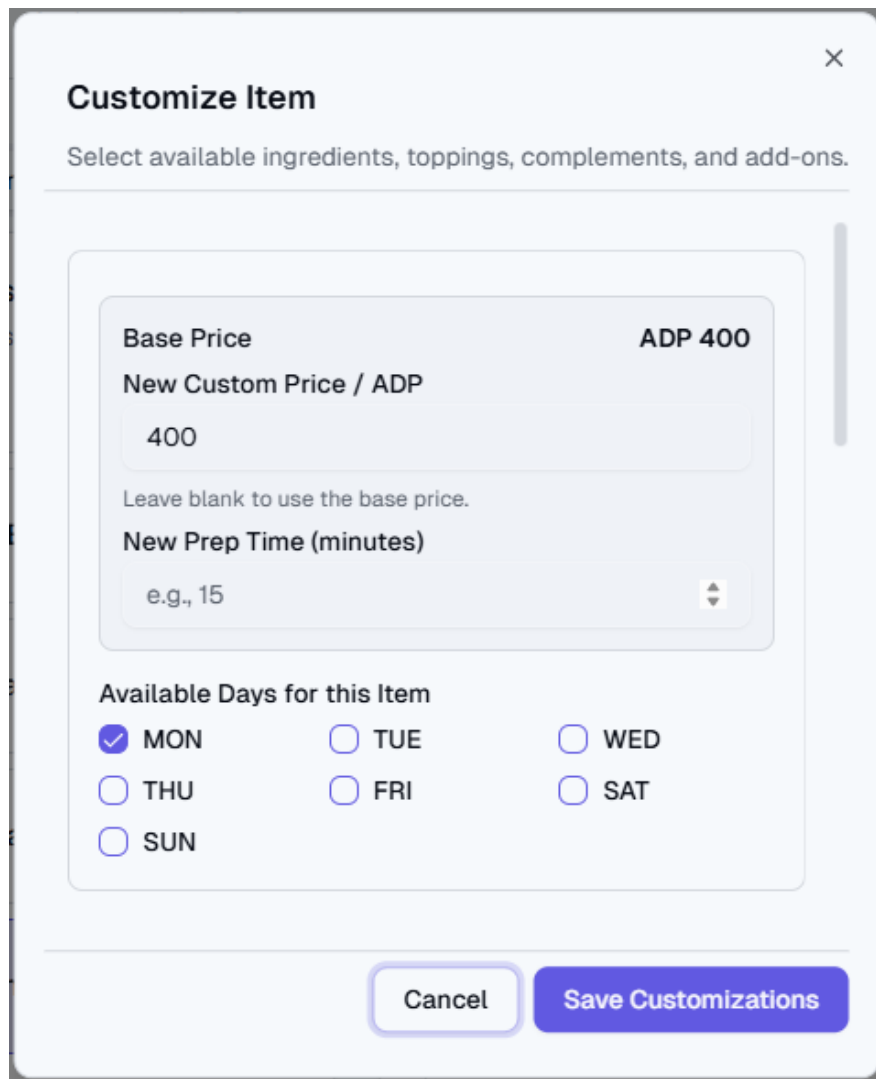


Figure 5: User Interface Screenshot Showing Menu Item Customization Dialog

2.5.1.2 Lodging Assignment Module

I created a floor-based accommodation management system where lodging businesses can organize their accommodations by floors. The module allows for creating and managing multiple floors, each containing various accommodation types. I implemented features for assigning accommodations to floors, setting room details such as maximum guests, bed configurations, amenities, pricing, and availability status

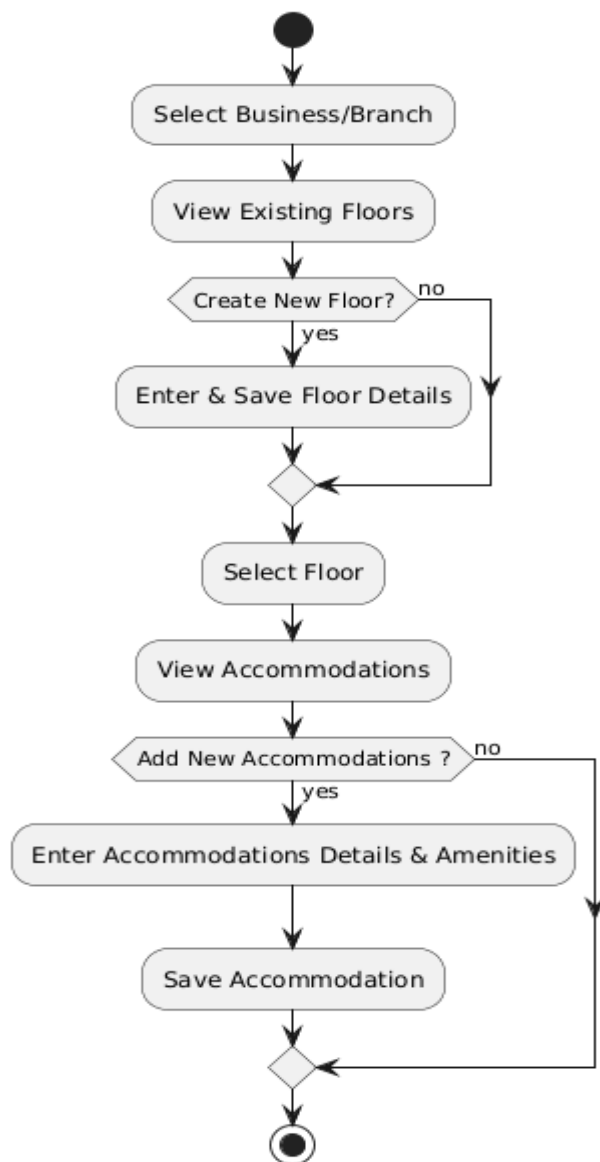


Figure 6: Activity Diagram Showing Lodging Accommodation Management Process

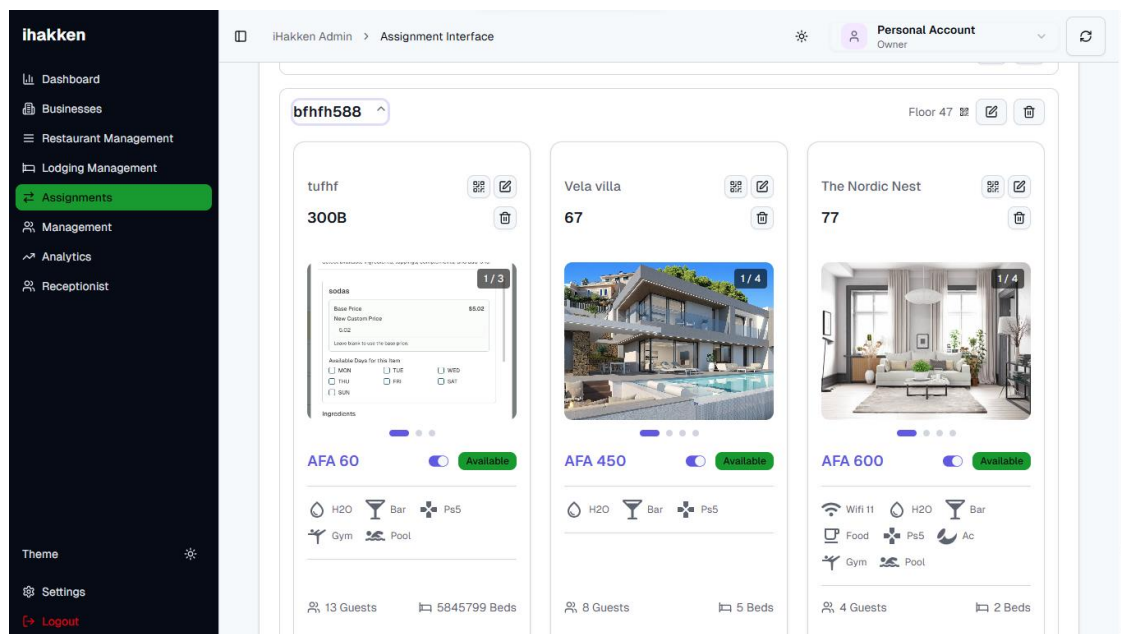
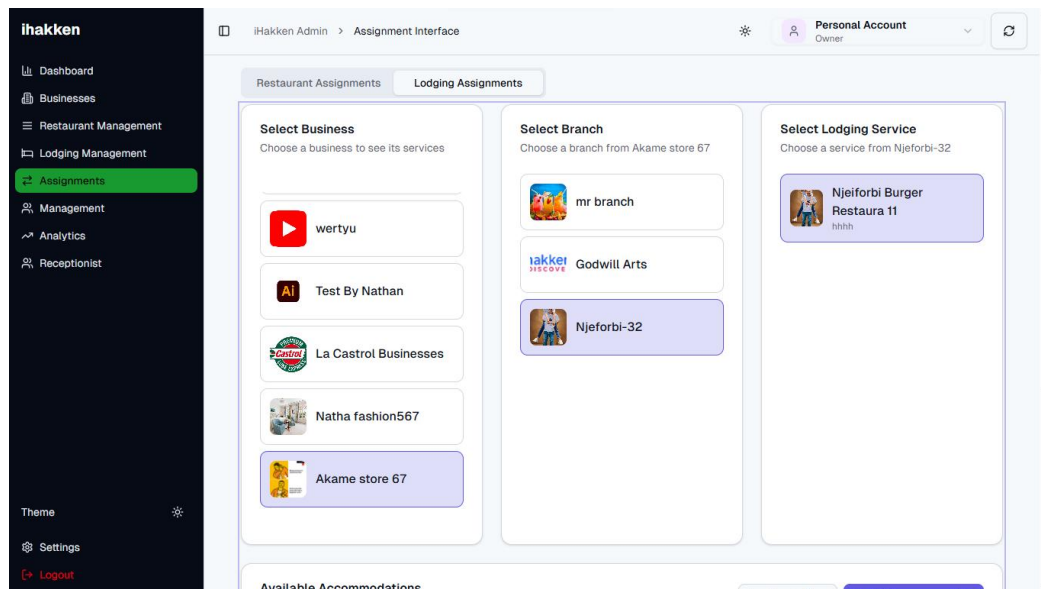


Figure 7: User Interface Screenshots Showing Lodging Assignment Interfaces

2.5.2 Analytics Dashboard

I developed a comprehensive analytics dashboard that provides insights into business performance. The dashboard includes:

- Key performance metrics (total QR scans, average time spent, active branches)
- Visual charts showing service distribution using pie charts
- Line charts displaying QR code scan trends
- Area charts for time spent analysis by service type
- Business performance tables with ranking and statistics
- Top performing items list combining both restaurant and lodging data

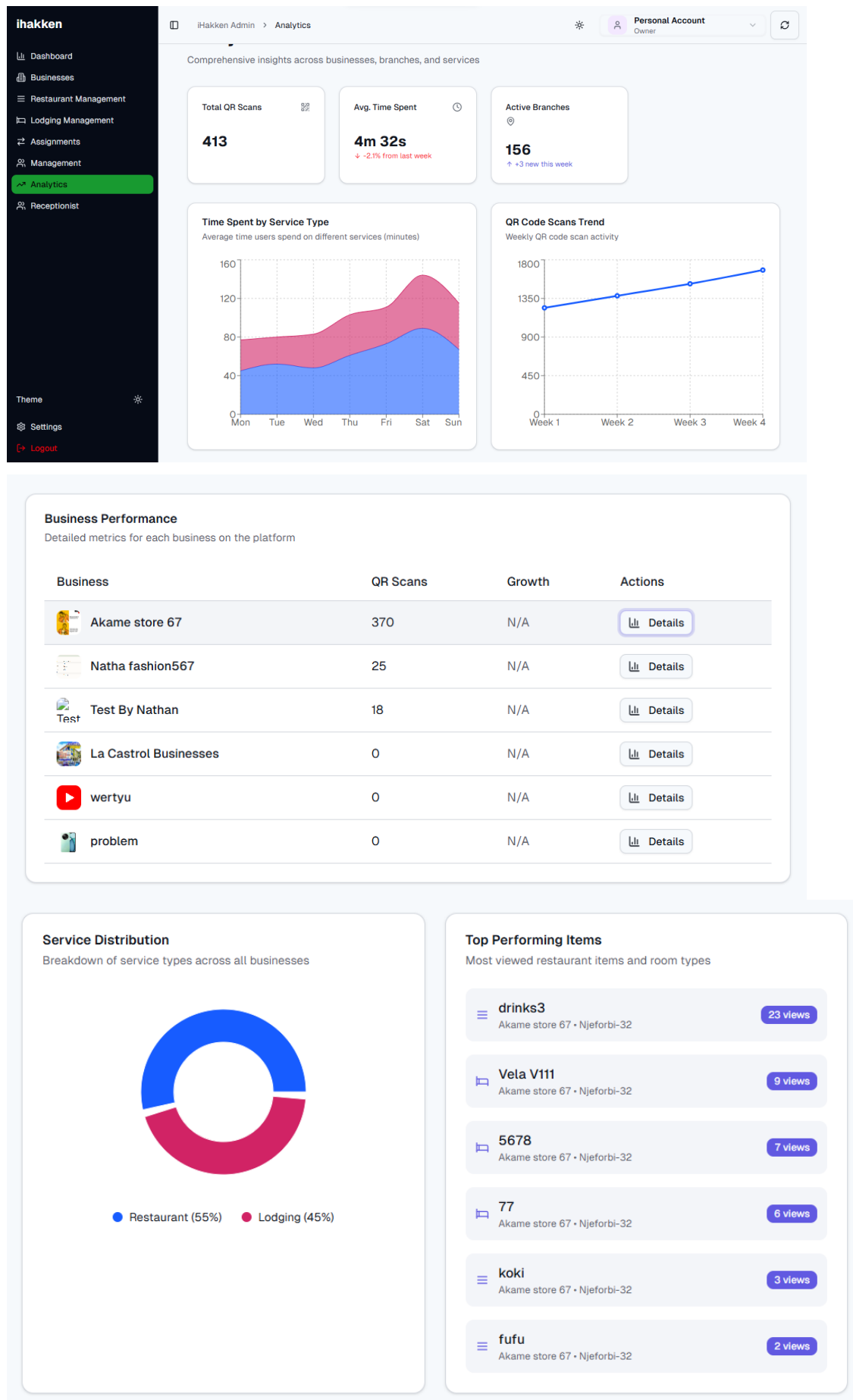


Figure 8: User Interface Screenshots of Analytics Dashboard Overview

2.5.3 QR Code Generation System

I created a QR code generation dialog that allows businesses to generate customized QR codes for their branches, services, tables, and accommodations. The system includes extensive customization options such as:

- Logo integration with size and style controls
- Color customization (foreground and background)
- Error correction levels
- Border and box size configurations
- Visual effects including rounded modules and glow effects
- Download functionality for generated QR codes

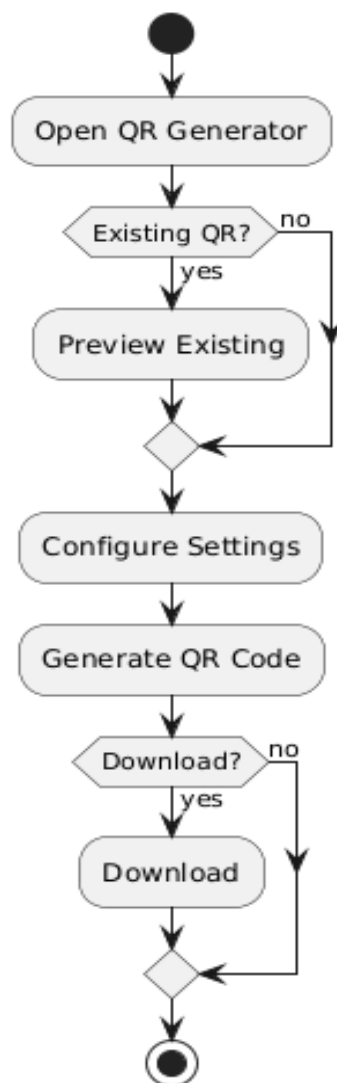


Figure 9: Activity Diagram for QR Code Generation Workflow

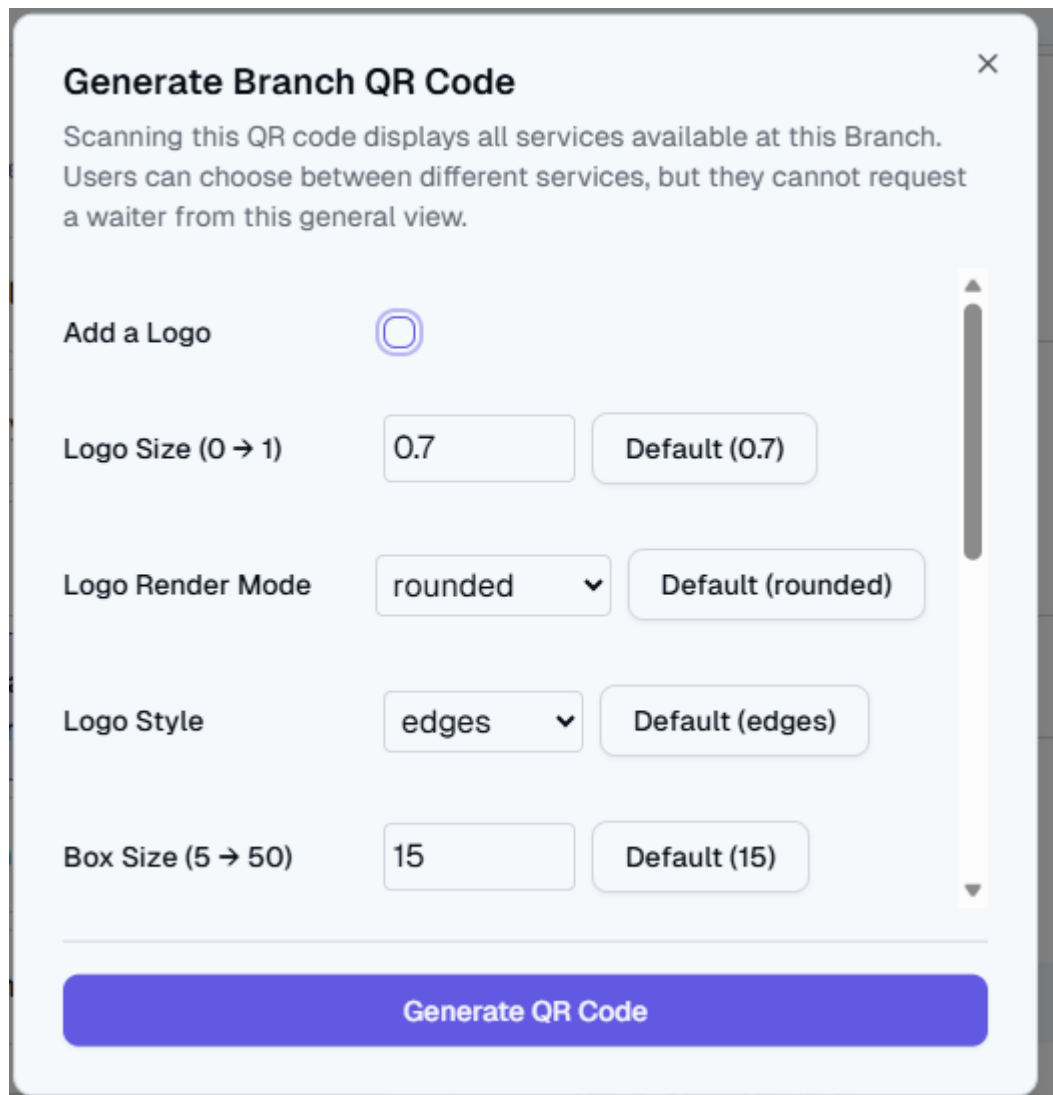


Figure 10: User Interface Screenshot for QR Code Generation Dialog

2.5.4 Service Management Interface

I developed a service card component that provides a tabbed interface for managing branch services. The component includes:

- Overview tab displaying service information and images
- Staff unit's management
- Table management for restaurant services
- Advertisement settings configuration
- AI Chabot configuration integration

2.5.5 Table Management System

I implemented a comprehensive table management system specifically designed for restaurant services. This module allows restaurant owners to:

- Create and manage tables within a restaurant service
- Assign unique names and labels to tables

- Edit table information including name and label
- Delete tables when no longer needed
- Generate QR codes for individual tables, enabling customers to scan and access table-specific services
- View all tables in an organized table format with action buttons

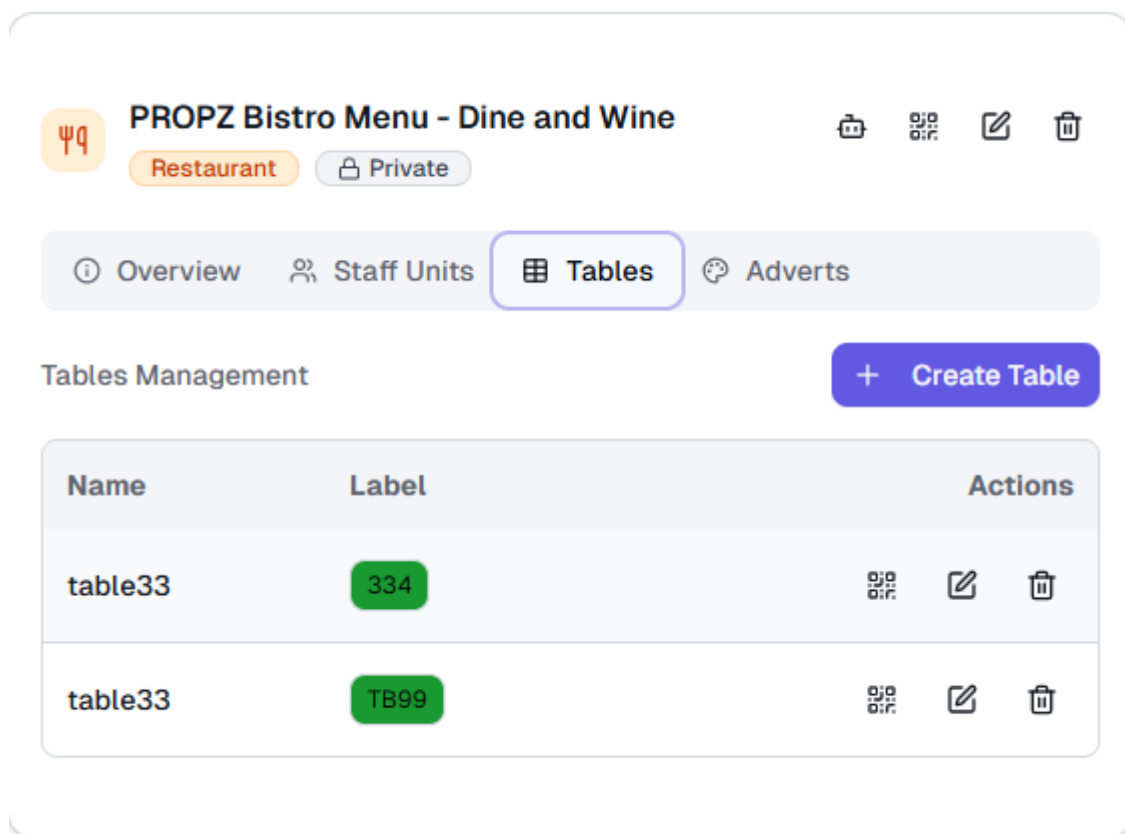


Figure 11: screenshot showing Tables that have been created for restaurants

2.5.6 Staff Units Management System

I developed a comprehensive staff unit's management system that enables businesses to organize and manage different staff roles or stations within their services. This system allows administrators to:

- Create staff units representing different roles
- Assign icons and descriptions to each staff unit for easy identification
- Manage staff unit status (active/inactive) through toggle switches
- Edit staff unit information including name, description, and icon
- Delete staff units when they are no longer needed

- Generate QR codes for staff units to link physical devices
- Configure device status and AI agent settings for each staff unit

The staff units system is particularly important for enabling device-based operations, where each staff unit can be linked to a physical device (such as a tablet or smartphone) that staff members use to interact with the system and respond to customer requests.

I implemented an advanced settings interface for staff units that include comprehensive device and AI agent configuration. This system provides:

Device Configuration:

- View linked device information and status
- Toggle device active/inactive status
- Delete device associations when needed
- Real-time device status updates

AI Agent Configuration:

- Enable or disable AI agents for staff units
- Configure AI agent name and identity
- Voice response configuration through:
 - In-browser voice recording with guided prompts
 - Audio file upload
 - Voice preview and playback functionality
 - Voice files management and replacement
- RAG (Retrieval-Augmented Generation) document management:
 - Upload documents for AI knowledge base
 - Manage and delete uploaded documents
 - View document library for each staff unit

The AI configuration system includes a sophisticated voice recording interface with pre-defined prompts to guide staff through recording professional greeting messages. The system supports both real-time recording using the browser's Media Recorder API and file uploads for pre-recorded audio files.

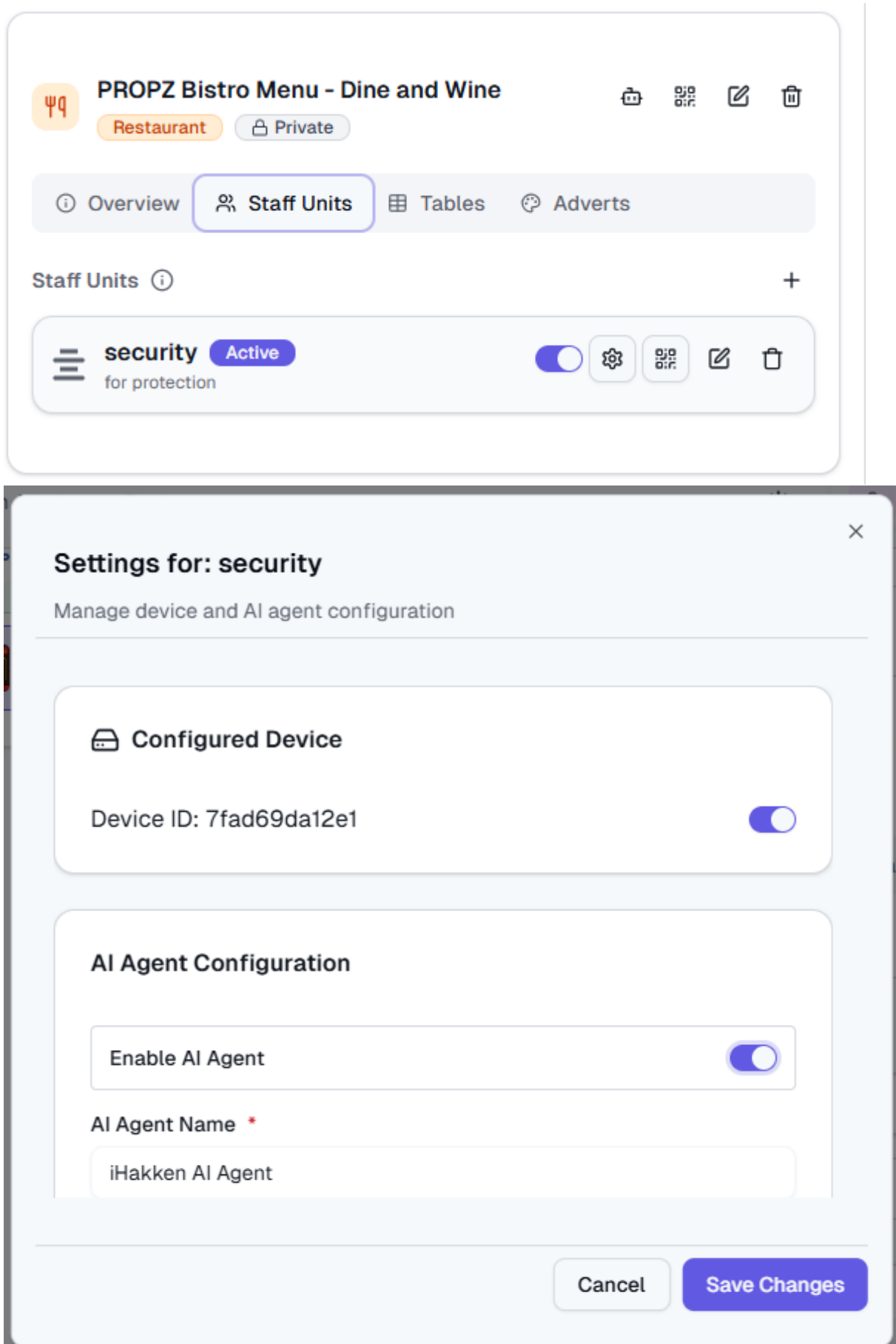


Figure 12: Screenshots of AI agent Configuration and the configured device

2.5.7 Settings Management Page

I created a comprehensive settings management interface with multiple tabs including:

- Profile management with image upload
- Account security and password change functionality
- Notification preferences
- Organization settings
- Currency selection

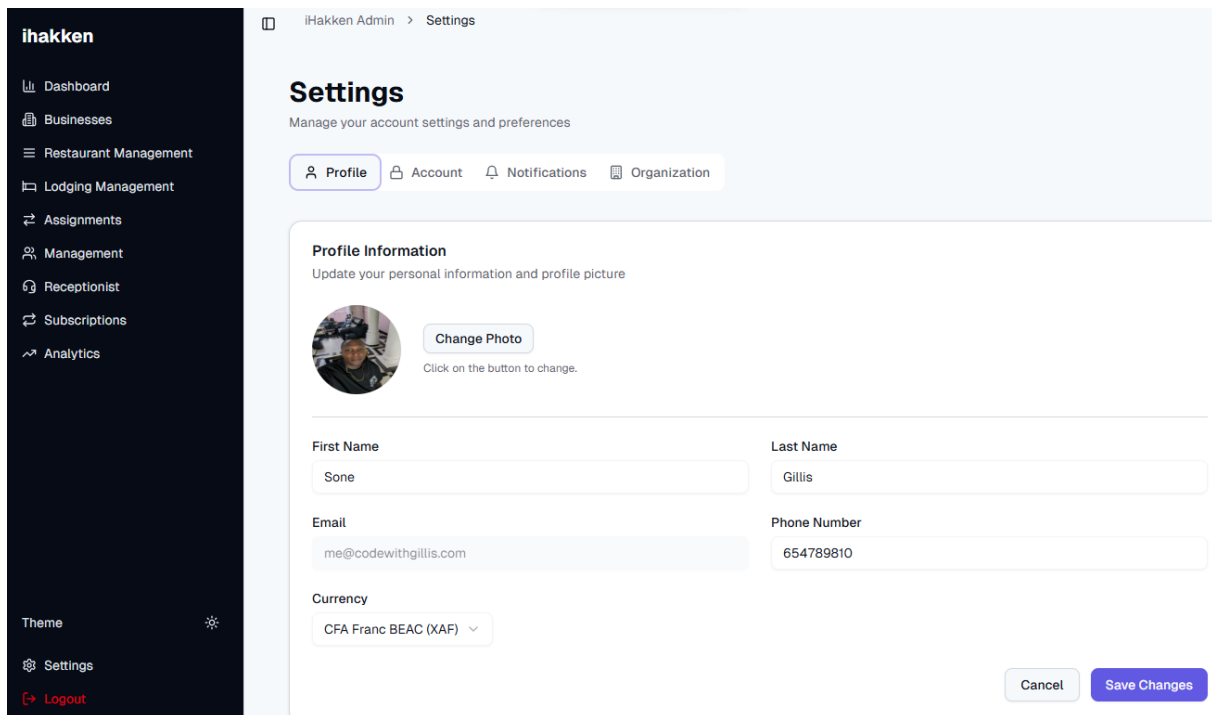
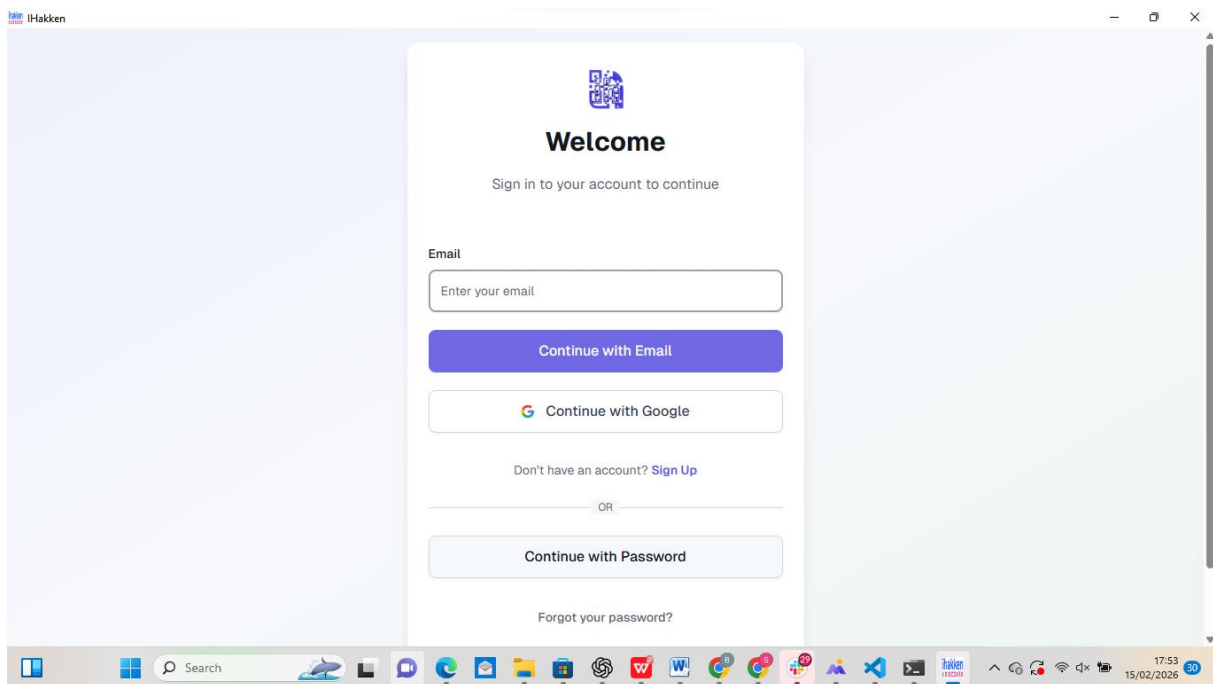


Figure 13: User Interface Screenshot showing Settings interface

2.6 Development of Native Desktop Application (Tauri Integration)

In addition to the web applications, I extended the iHakken ecosystem by developing a native desktop application using the Tauri framework. By leveraging the system's native web view and a Rust-based backend, I was able to create a lightweight, high-performance "Remote Wrapper" for the production site. This approach significantly improved the user experience for business administrators by providing a dedicated, standalone tool that bypasses the limitations of a standard browser tab.

To ensure a seamless transition for desktop users, I specifically configured the application to route directly to the Production Authentication environment (<https://ihakken.com/auth>). I further enhanced the application's visual identity by implementing a suite of custom anime-themed icons and professional window properties, such as resizable and optimized default dimensions. This ensured that the iHakken branding remained consistent and visually engaging across both the Windows and macOS platforms.



2.7 DevOps and Continuous Integration/Continuous Deployment (CI/CD)

During the internship, I was introduced to DevOps practices, specifically Continuous Integration and Continuous Deployment (CI/CD) using GitHub Actions. I was actively involved in writing and configuring deployment workflows for the Ihakken project, covering both the **User-facing application** and the **Admin Management application**. I implemented GitHub workflow pipelines that automatically deploy these applications to a **Contabo Virtual Private Server (VPS)** whenever changes are pushed to the main branch of the project repository. I specifically configured these pipelines to manage sophisticated dual environment architecture, ensuring that the **User App** and the **Admin App** were correctly routed to their respective Production and Development URLs:

- **Production Environment:**
 - User App: Hosted at **ihakken.com**
 - Admin App: Hosted at **app.ihakken.com**
- **Development Environment:**
 - User App: Hosted at **app.dev.ihakken.com**
 - Admin App: Hosted at **dev.ihakken.com**

These workflows handled tasks such as source code checkout, file synchronization to the exact directories on the VPS, Docker image building, and restarting application services using Docker Compose. By using a strict sequence of commands and environment specific override

files, I ensured that the configuration for the User App remained isolated from the Admin App, maintaining system stability across all subdomains.

This automation significantly improved deployment efficiency by reducing manual intervention, minimizing deployment errors, and ensuring consistency across environments. Through this experience, I gained practical knowledge of CI/CD pipelines, GitHub Actions, Docker, and the technical complexities of managing a multi-subdomain server-based application deployment.

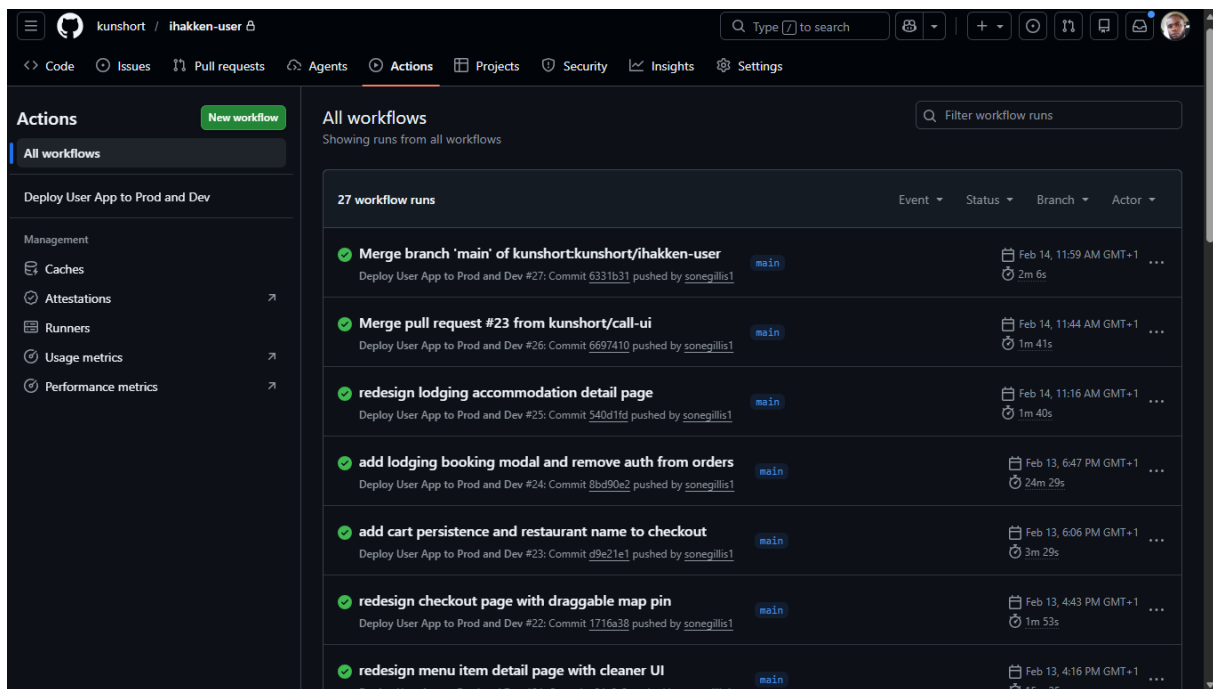


Figure 14: screenshot of a successful deployment in the ihakken user application

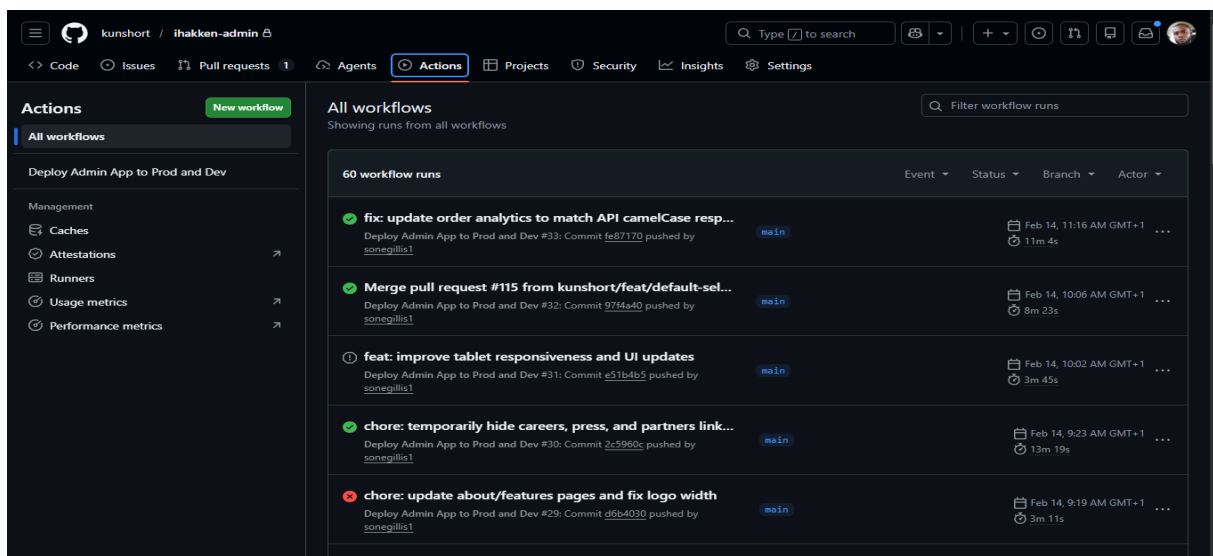


Figure 15: screenshot of a successful deployment in the ihakken admin application

2.8 User Facing Application

2.8.1 Request Waiter Feature

I implemented a Call Waiter feature within the restaurant interface that allows customers to request assistance directly from their table without needing to flag down staff manually. I integrated a prominent notification button into the sticky header for easy access. The system identifies the customer's specific table and sends a request to the staff. I also designed the user feedback mechanism, which displays immediate visual confirmation (toast notifications) to let the customer know their request has been successfully sent.

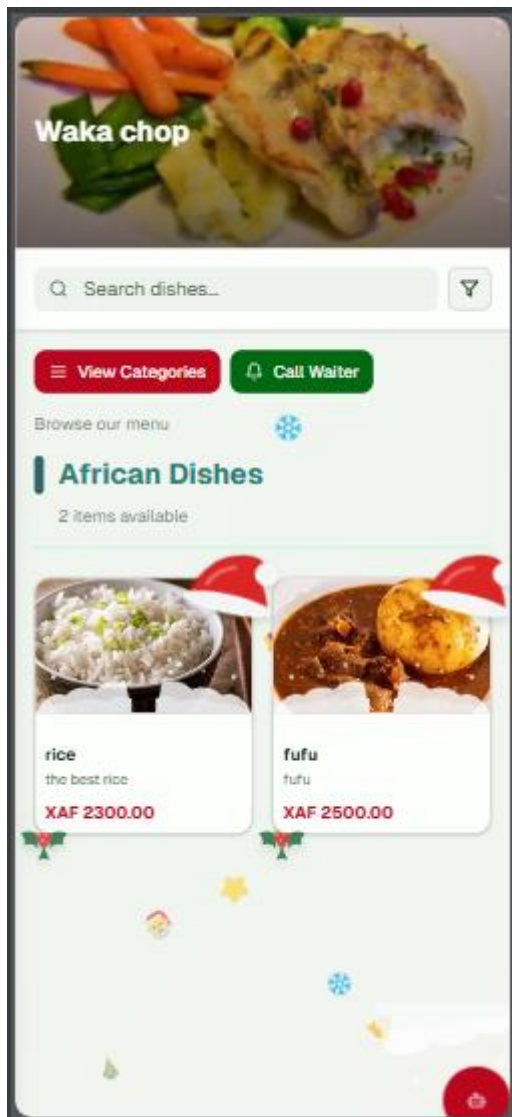


Figure 16

Figure 16: Screenshot showing the call waiter button

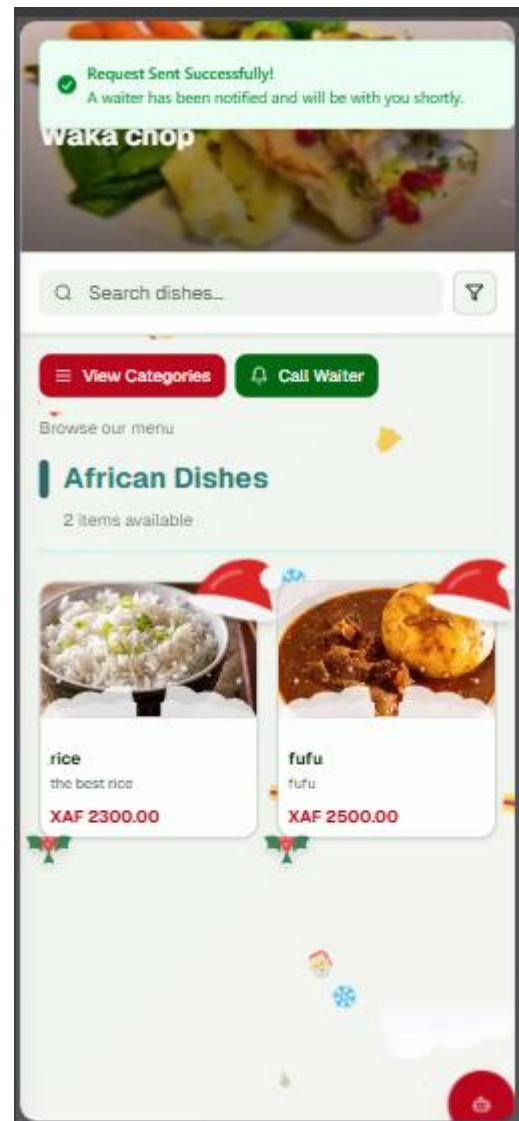


figure 17

Figure 17: Screenshot showing after the user has click the call waiter button

2.8.2 AI Chat Assistant Module

I developed a comprehensive, interactive AI Chat Assistant available across both the Restaurant and Lodging sections of the application. This module provides users with a real-time conversational interface to ask questions or get recommendation.

- **Interactive Interface:** I built a collapsible chat window with a floating action button, featuring a modern design with typing indicators and auto scrolling to ensure a smooth user experience.
- **Rich Media Responses:** Beyond simple text, I enabled the chat to display dynamic content cards, such as specific menu items details directly within the conversation flow, allowing users to view prices and images instantly.
- **Session Continuity:** I implemented session management that preserves the chat history, ensuring that the conversation context is maintained even if the user refreshes the page or navigates between different services.

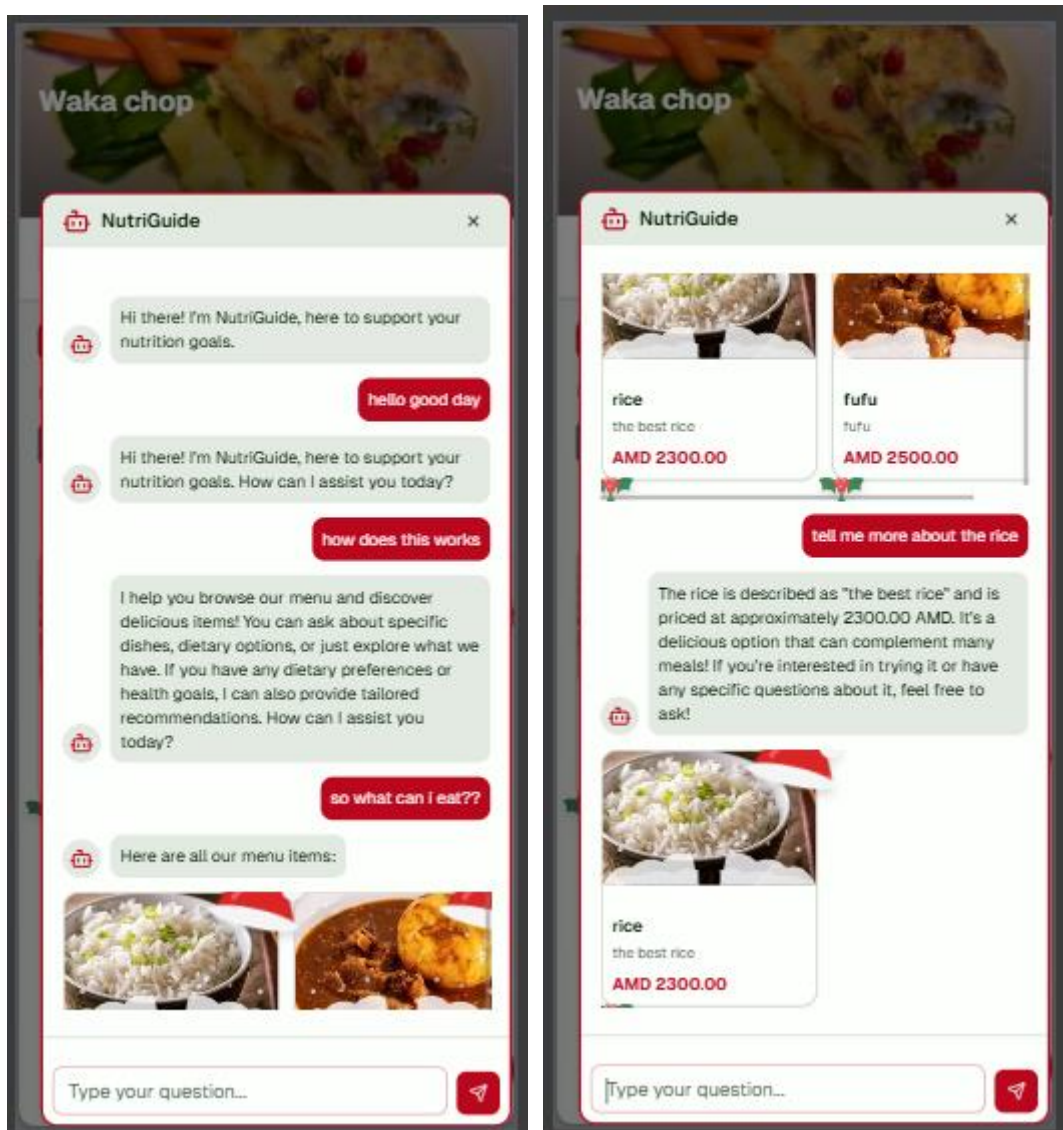


Figure 18: screenshots showing how a user is chatting with the AI Chabot

2.9 How I Performed the Tasks

2.9.1 Technical Approach and Methodology

Frontend Framework and Libraries: I utilized React with TypeScript as the primary frontend framework, leveraging modern React hooks such as `useState`, `useEffect`, and `useMemo` for state management and performance optimization. I employed the TanStack Form library (formerly React Hook Form) for robust form handling with validation, which was essential for the complex forms in the assignment and customization modules.

State Management: For API communication and server state management, I implemented Redux Toolkit Query (RTK Query), which provided efficient data fetching, caching, and automatic refetching capabilities. This was particularly important for handling the hierarchical data structure and ensuring data consistency across the application.

UI Component Library: I used a component library based on Radix UI primitives, which provided accessible and customizable UI components. I customized these components to match the application's design system, ensuring consistent styling and behavior across all modules.

2.9.2 Development Process

Component Architecture: I followed modular component architecture, creating reusable components that could be shared across different features. For instance, the `BusinessSelector` component was designed to work with both restaurant and lodging assignment modules. I also created shared components for loading states, error handling, and empty states to maintain consistency throughout the application.

API Integration: I integrated with the backend REST API using RTK Query mutations and queries. I handled various API response scenarios including success, error, and loading states. I implemented proper error handling with user-friendly error messages displayed through toast notifications. For file uploads (images, logos), I utilized `FormData` to send multipart/form-data requests.

Data Flow Management: I implemented a cascading data loading approach where selecting a business would trigger branch loading, which in turn would trigger service loading. I used query skip conditions and conditional fetching to optimize API calls and prevent unnecessary requests. This approach was crucial for the assignment interfaces where multiple levels of selection were required.

Real-time Updates: I implemented automatic cache invalidation and refetching after mutations to ensure the UI always displayed up-to-date information. For example, after creating a floor, the floors list would automatically update without requiring manual refresh.

2.9.3 User Experience Enhancements

Loading States: I implemented comprehensive loading indicators using skeleton screens and loading spinners to provide visual feedback during data fetching operations. This was especially important for the analytics dashboard where multiple API calls were happening simultaneously.

Error Handling: I created error boundary components and error display components that provided clear error messages and retry functionality. For network errors, I implemented retry mechanisms with user feedback.

Search and Filtering: I implemented client-side search and filtering functionality for accommodations and menu items. The search feature allowed users to quickly locate specific items by code or name, improving efficiency in managing large datasets.

CHAPTER 3.Extra-Curricular Activities

3.1 Introduction

During the internship period at Kunshort Company Ltd, activities were not strictly limited to office-based software development. I also had the opportunity to participate in activities that complemented the technical training and provided a broader professional experience. In addition to the daily development tasks, I took part in extracurricular activities organized during the internship period. These activities allowed us to relax, interact freely with supervisors and team members, and strengthen professional relationships outside the normal working environment. The main extracurricular activities during my internship were the weekly presentations and a team outing to Seme Beach in Limbe.

3.2 Weekly Presentations

During the internship period, we held weekly presentations where each intern presented the work completed during the week. In these sessions, we explained the progress made on the project, the specific tasks we handled, and the challenges encountered while working on them.

These presentations were important because they helped to track the overall progress of the project. They also allowed the supervisors to evaluate our performance and guide us where necessary. When corrections were given, we took note of them and improved our work in the following week.

The weekly presentations also helped me to improve my confidence in explaining technical concepts in front of others. Standing before colleagues and supervisors to present my work strengthened my communication skills and helped me better understand the parts of the project I was working on.

3.3 Trip to Seme Beach, Limbe

One of the extracurricular activities carried out during the internship period was a team outing to Seme Beach in Limbe. Seme Beach is located along the coastal area of Limbe in the South West Region of Cameroon. It is situated by the Atlantic Ocean and is known for its calm environment and natural beach setting, which makes it a suitable place for relaxation and group activities.

During the outing, we spent time walking around the beach, enjoying the fresh air and the ocean view. Some of us swam, while others relaxed and interacted freely. We took several

photos together to keep memories of the day. The atmosphere was friendly and informal, which allowed everyone to feel relaxed outside the normal office environment.

After spending time at the beach, we went to a restaurant where we ate and had drinks together. While at the restaurant, we had meaningful discussions with our supervisors and members of the technical team. They shared advice with us about life after school and what to expect in the professional world.

Based on their observations of us throughout the six-month internship period, the supervisors spoke to each intern about their personal strengths and weaknesses. They advised us on the areas where we needed improvement and encouraged us to maintain the areas where we were performing well. Their guidance was practical and realistic, and it gave us a clearer understanding of how to grow both professionally and personally.

The outing was both enjoyable and educational. It strengthened the bond between interns and supervisors and made the internship experience more balanced and memorable.



Figure 19: Picture showing interns and the staffs of Kunshort Company at Seme beach in Limbe



Figure 20: picture showing interns and one staff member after swimming

CHAPTER 4: DIFFICULTIES AND CHALLENGES ENCOUNTERED

During the internship period at Kunshort Company Ltd, several difficulties and challenges were encountered. These challenges were technical, environmental, and logistical in nature. Despite these difficulties, appropriate strategies were adopted to minimize their impact and ensure the successful completion of assigned tasks. The challenges encountered and the strategies used to overcome them are discussed below.

4.1 Difficulty in Understanding React and Next.js

One of the major challenges encountered during the internship understood React and Next.js, which were the main frontend frameworks used in developing the Ihakken platform. At the initial stage, concepts such as component-based architecture, React hooks, routing, and server-side rendering were difficult to fully understand. This made it challenging to implement some frontend features efficiently.

To overcome this challenge, additional personal time was dedicated to learning. I watched several YouTube tutorial videos, followed online practical demonstrations, and practiced by building and modifying small components within the project. Continuous practice and exposure gradually improved my understanding of React and Next.js, enabling me to contribute effectively to the development process.

4.2 Difficulty in Understanding a Large Codebase

The Ihakken platform consisted of a large and complex codebase, which made it difficult at first to understand how different modules and components interacted with each other. Navigating through multiple files and directories initially slowed down the development process.

To overcome this challenge, I studied the project structure carefully, traced data flow between components, and reviewed comments and documentation within the codebase. I also sought clarification from my professional supervisor when necessary. This approach improved my ability to work effectively within large-scale software projects.

4.3 Power Supply and Network Connectivity Challenges

Irregular power supply (light issues) was another major challenge encountered during the internship. Power outages often resulted in slow or unstable internet connectivity, which

affected activities such as downloading dependencies, syncing code repositories, and accessing online resources. This situation slowed down the overall working process.

To manage this challenge, I planned tasks that could be done offline during power outages, such as writing code, debugging, and reviewing documentation. When internet access was available, pending tasks were completed and synchronized. This helped reduce productivity loss.

4.5 Transportation Challenges

Transportation to the internship office also posed a challenge. Due to distance, transport availability, and cost, commuting to the workplace was sometimes difficult and time-consuming. This occasionally affected punctuality and increased physical fatigue.

To mitigate this challenge, I planned my movements in advance, left home earlier when necessary, and adjusted my daily schedule to ensure timely arrival at the workplace.

4.6 Challenges Related to the Anglophone Crisis

Another significant challenge encountered during the internship was the impact of the Anglophone crisis, which resulted in a general lockdown lasting approximately two weeks. During this period, movement was restricted, making it impossible to physically report to the internship office.

This situation disrupted normal work activities and slowed overall progress. Although remote work was attempted, limited internet connectivity reduced productivity. To overcome this challenge, I focused on self-learning, reviewing project documentation, and preparing for task execution once normal operations resumed. After the lockdown, I returned to the workplace and continued my assigned duties.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The internship carried out at Kunshort Company Ltd was a highly valuable and enriching experience that provided practical exposure to real-world software engineering practices. The internship successfully bridged the gap between theoretical knowledge acquired in the classroom and practical application in a professional work environment.

During the internship period, I gained hands-on experience in modern frontend development using technologies such as React, Next.js, TypeScript, and Redux Toolkit Query. In addition to frontend development, I was also exposed to DevOps practices, particularly Continuous Integration and Continuous Deployment (CI/CD) using GitHub Actions, Docker, and server deployment workflows. I actively participated in writing deployment pipelines that automated the deployment of company projects to a Virtual Private Server, thereby improving efficiency and reliability.

Despite encountering several challenges such as difficulties in learning new technologies, power supply and network issues, transportation challenges, and disruptions caused by lockdowns during the election period and the Anglophone crisis, appropriate strategies were adopted to overcome these obstacles. Overall, the internship greatly improved my technical competence, adaptability, and confidence, and prepared me for future professional responsibilities in the field of software engineering.

5.2 Recommendations

5.2.1 Recommendations to the Company

- The company should continue strengthening remote work structures during periods of restricted movement by providing clear communication channels and flexible work arrangements.
- Investment in reliable power backup systems and stable internet connectivity would help minimize disruptions caused by power outages and network instability.

5.2.2 Recommendations to the School

- The university should encourage early exposure to modern frameworks such as React and Next.js before students begin their internship period. This will reduce the learning curve and improve productivity during industrial training.
- More practical-oriented courses and workshops should be organized to better prepare students for real-world software development environments.

5.2.3 Recommendations to the Society

- There should be efforts to improve infrastructure such as stable electricity supply and internet services to support students and organizations in the technology sector.
- A stable and peaceful environment should be promoted to prevent disruptions such as lockdowns, which negatively affect academic and professional activities.

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APPENDICES

This section contains supplementary documents and materials that support and provide evidence for the internship activities described in this report. These appendices include official documents, progress records, and visual proof of practical work carried out during the internship period at Kunshort Company Ltd.

APPENDIX A: Internship Posting Letter

This appendix presents a copy of the **internship posting letter** issued by the **Faculty of Engineering and Technology, University of Buea**, authorizing the internship at Kunshort Company Ltd as part of the academic requirements for the Bachelor of Engineering degree.

UNIVERSITY OF BUEA
P.O. Box 63,
Buea, South West Region
CAMEROON
Tel: (237) 3332 21 34/3332 26 90/3332 27 06/3332 28 13
Fax: (237) 3332 22 77

REPUBLIC OF CAMEROON
PEACE - WORK - FATHERLAND

FACULTY OF ENGINEERING AND TECHNOLOGY

Dean: Agbor Dieudonne Agbor, PhD
V/DPA: Dr. Nde D. Nguti
V/DPA: Ngeahli Divine Khan, PhD
V/DRC: Tsafack Pierre, PhD
Faculty Officer: Nyeka-Sanga nee Emeli Akanang Eanice

Date: 03 APR 2025

Letter of Recommendation

Office of Human Resources

Mr. ABANDA SERGIO ABANDA is a student of the Faculty of Engineering and Technology (FET) of the University of Buea, with matriculation number FE22A131. He is currently a Level 400 student studying for an undergraduate degree programme in **Computer Engineering**. Admissions into this Faculty is based on academic excellence, consequently only meritorious candidates are enrolled into the Faculty. Being a Level 400 student, the candidate must have acquired some basic academic knowledge and skills necessary to undertake such practicum.

We so recommend Mr. ABANDA SERGIO ABANDA for a position of internship in your Organization in any Engineering duties and research activities of his field. This placement will enhance the professional development of the student and we count on his sense of commitment and assiduity to ensure a successful training.

Sincerely,

Marie Ebob Agboraboh Njoug, PhD
Associate Professor of Microbiology
FOS Internships
P.O. Box 63, University of Buea, Buea, Cameroon
Cell: +237 675301641, +237 605791827
Email: agboraboh.marie@unibuea.net, marieebob@yahoo.com

Figure 21: Internship posting form

APPENDIX B: Daily and Weekly Internship Reports

This appendix contains selected daily and weekly internship progress reports compiled during the internship period. These reports summarize tasks performed, technologies used, challenges encountered, and lessons learned throughout the training.

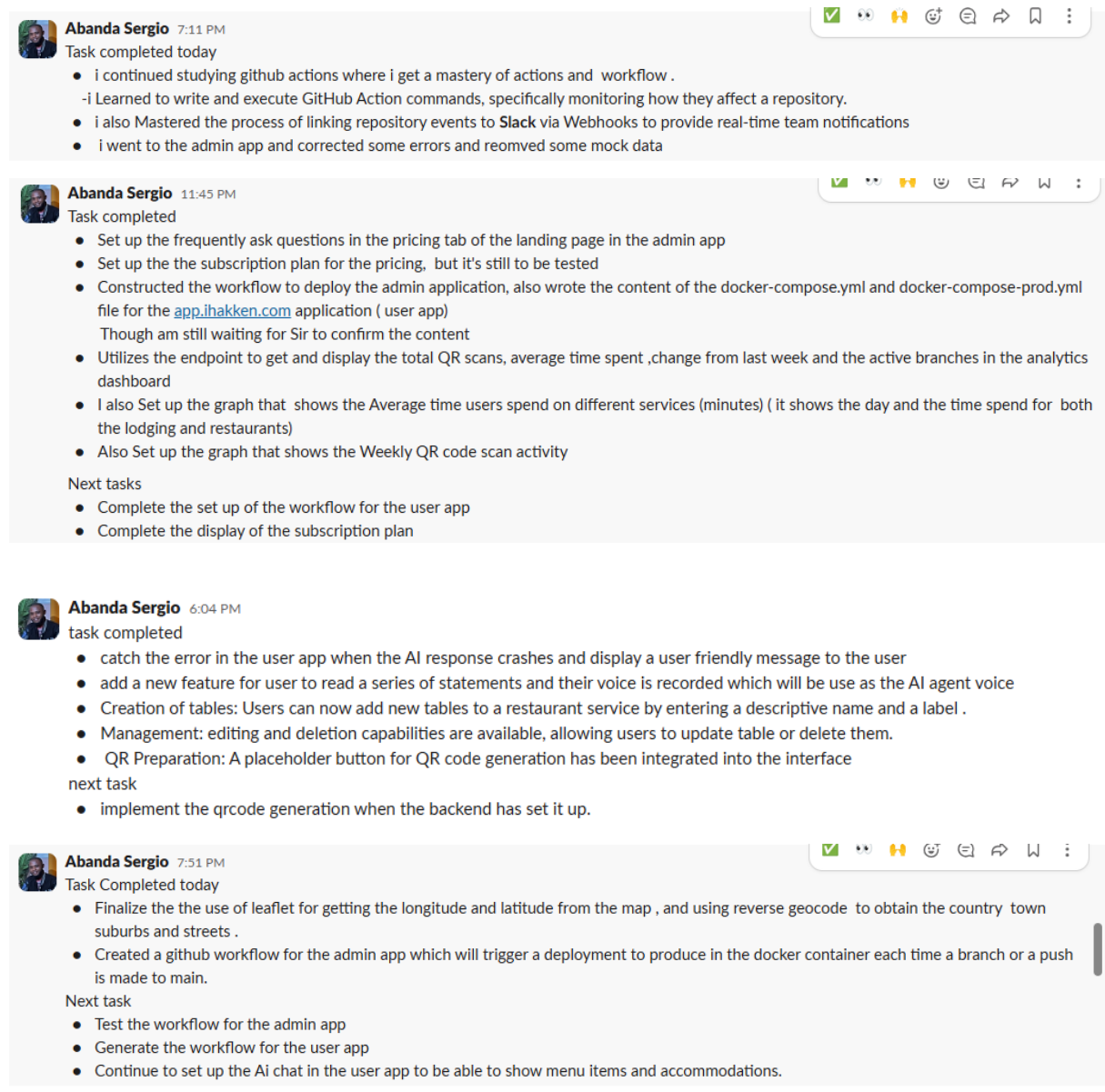


Figure 22: Screenshot's showing my daily task completed in slack

APPENDIX C: Tools and Technologies Used

This appendix provides a summarized list of the major tools, frameworks, and technologies utilized during the internship, as discussed in the report:

- React and Next.js
- TypeScript and JavaScript

- Redux Toolkit Query (RTK Query)
- Docker
- GitHub and GitHub Actions
- PostgreSQL
- Python and Django